

DSBD skill 3.1

Code:

```
CREATE DATABASE bookflow_db;
```

```
USE bookflow_db;
```

```
-- =====
```

```
-- JOIN OPERATIONS
```

```
-- =====
```

```
CREATE TABLE class (
```

```
    id INT,
```

```
    name VARCHAR(30)
```

```
);
```

```
CREATE TABLE class_info (
```

```
    id INT,
```

```
    address VARCHAR(30)
```

```
);
```

```
INSERT INTO class VALUES
```

```
(1, 'abhi'),
```

```
(2, 'adam'),
```

```
(4, 'alex');
```

```
INSERT INTO class_info VALUES
```

```
(1, 'DELHI'),
```

```
(2, 'MUMBAI'),
```

```
(3, 'CHENNAI');
```

```
-- CROSS JOIN
```

```
SELECT *
```

```
FROM class
```

```
CROSS JOIN class_info;
```

```
-- Recreate class table with additional data
```

```
DROP TABLE class;
```

```
CREATE TABLE class (
```

```
    id INT,
```

```
    name VARCHAR(30)
```

```
);
```

```
INSERT INTO class VALUES
```

```
(1, 'abhi'),
```

```
(2, 'adam'),
```

```
(3, 'alex'),
```

```
(4, 'anu');
```

```
-- INNER JOIN
```

```
SELECT *
```

```
FROM class
```

```
INNER JOIN class_info
```

```
ON class.id = class_info.id;
```

```
-- INNER JOIN with selected columns
```

```
SELECT class.name, class_info.address
```

```
FROM class
```

```
INNER JOIN class_info
```

```
ON class.id = class_info.id;
```

```
-- NATURAL JOIN
```

```
SELECT *
```

```
FROM class
```

```
NATURAL JOIN class_info;
```

```
-- Add more unmatched records
```

```
INSERT INTO class VALUES
```

```
(5, 'ashish');
```

```
INSERT INTO class_info VALUES
```

```
(7, 'NOIDA'),
```

```
(8, 'PANIPAT');
```

```
-- LEFT OUTER JOIN
```

```
SELECT *
```

```
FROM class
```

```
LEFT OUTER JOIN class_info
ON class.id = class_info.id;
```

```
-- LEFT JOIN - unmatched records from class
```

```
SELECT *
FROM class
LEFT JOIN class_info
ON class.id = class_info.id
WHERE class_info.id IS NULL;
```

```
-- RIGHT OUTER JOIN
```

```
SELECT *
FROM class
RIGHT OUTER JOIN class_info
ON class.id = class_info.id;
```

```
-- RIGHT JOIN - unmatched records from class_info
```

```
SELECT *
FROM class
RIGHT JOIN class_info
ON class.id = class_info.id
WHERE class.id IS NULL;
```

```
-- FULL OUTER JOIN using UNION
```

```
SELECT *  
FROM class  
LEFT JOIN class_info  
ON class.id = class_info.id
```

UNION

```
SELECT *  
FROM class  
RIGHT JOIN class_info  
ON class.id = class_info.id;
```

-- FULL OUTER JOIN - unmatched records only

```
SELECT *  
FROM class  
LEFT JOIN class_info  
ON class.id = class_info.id  
WHERE class_info.id IS NULL
```

UNION

```
SELECT *  
FROM class  
RIGHT JOIN class_info  
ON class.id = class_info.id  
WHERE class.id IS NULL;
```

```
-- =====
```

```
-- SET OPERATORS
```

```
-- =====
```

```
CREATE TABLE first_table (
```

```
    id INT,
```

```
    name VARCHAR(30)
```

```
);
```

```
CREATE TABLE second_table (
```

```
    id INT,
```

```
    name VARCHAR(30)
```

```
);
```

```
INSERT INTO first_table VALUES
```

```
(1, 'abhi'),
```

```
(2, 'adam');
```

```
INSERT INTO second_table VALUES
```

```
(2, 'adam'),
```

```
(3, 'chester');
```

```
-- UNION
```

```
SELECT *
```

```
FROM first_table
```

UNION

SELECT *

FROM second_table;

-- UNION with selected column

SELECT name

FROM first_table

UNION

SELECT name

FROM second_table;

-- UNION ALL

SELECT *

FROM first_table

UNION ALL

SELECT *

FROM second_table;

-- COUNT rows using UNION ALL

SELECT COUNT(*)

```
FROM (  
    SELECT *  
    FROM first_table  
  
    UNION ALL  
  
    SELECT *  
    FROM second_table  
) AS A;
```

```
-- INTERSECT  
SELECT *  
FROM first_table
```

```
INTERSECT
```

```
SELECT *  
FROM second_table;
```

```
-- EXCEPT (equivalent of MINUS)  
SELECT *  
FROM first_table
```

```
EXCEPT
```

```
SELECT *
```

```
FROM second_table;
```